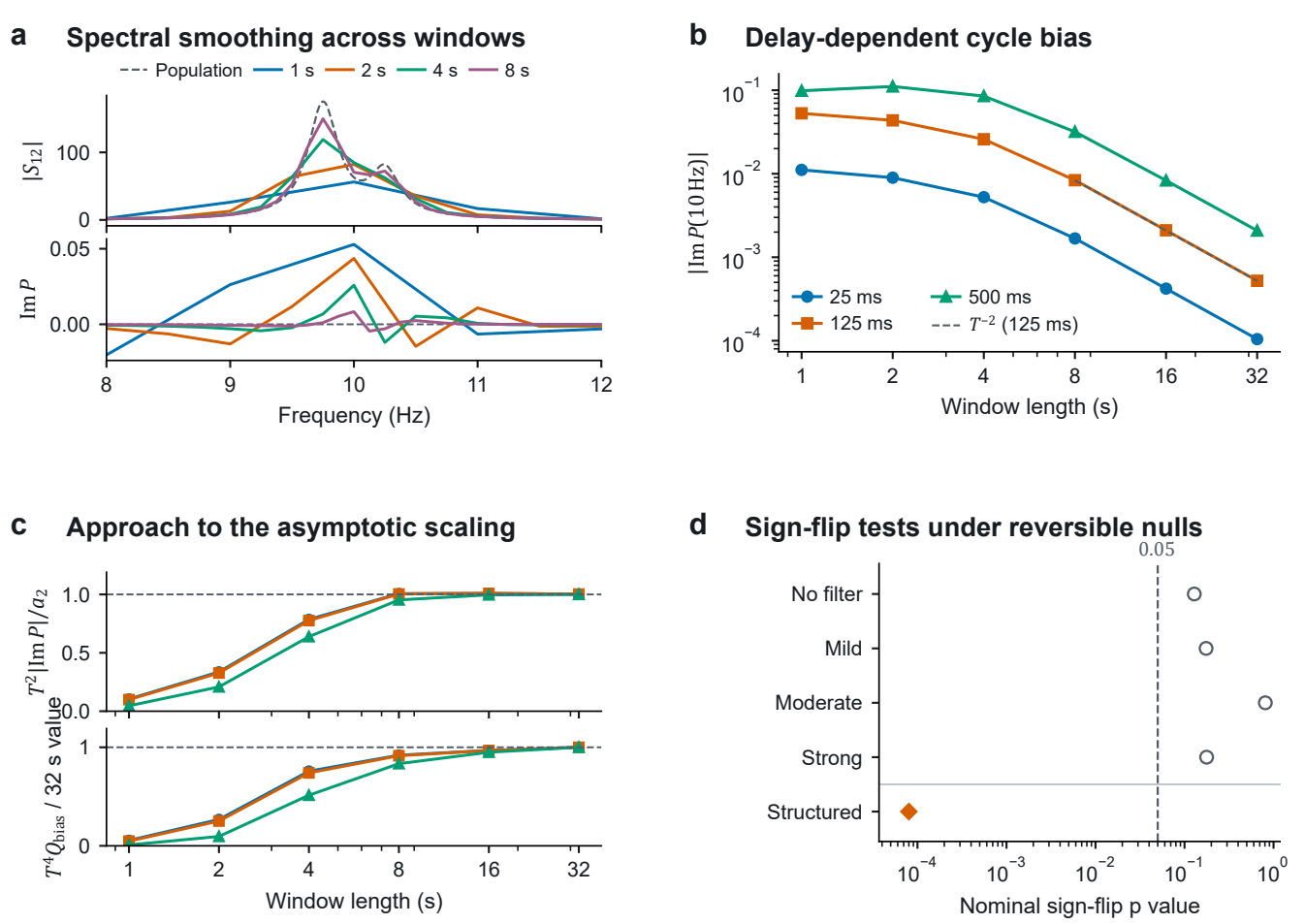

a
Spectral smoothing across windows
Population
1 s
2 s
4 s
8 s
$|S_{12}|$
100
0
0.05
Im P
0.00
8
9
10
11
12
Frequency (Hz)
b
Delay-dependent cycle bias
$|\mathrm{Im}\,P(10\,\mathrm{Hz})|$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
25 ms
500 ms
125 ms
T^{-2} (125 ms)
1
2
4
8
16
32
Window length (s)
c
Approach to the asymptotic scaling
$T^2|\mathrm{Im}\,P|/a_2$
1.0
0.5
0.0
T^4Q_{bias} / 32 s value
1
0
1
2
4
8
16
32
Window length (s)
d
Sign-flip tests under reversible nulls
0.05
No filter
Mild
Moderate
Strong
Structured
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Nominal sign-flip p value